Selective Alignment Transformer for Partial-Set Remote Sensing Image Cross-Scene Classification

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***Abstract*— Cross-scene classification aims to transfer knowledge acquired from a label-rich source domain to an unlabeled target domain with a distribution shift. With the large amount of available remote sensing data from diverse satellite platforms, it prompts us to utilize the knowledge from extensive datasets to address target tasks in small-scale domains, known as partial domain adaptation (PDA). However, the PDA setting poses significant challenges for remote sensing scene images. Existing methods often fail to sufficiently explore both task-specific and transferable knowledge across domains based on the representations of entire samples, potentially resulting in the amplification of negative transfer brought by irrelevant knowledge. To address this, we propose a new selective alignment transformer (SAT) designed to distinguish transferable and untransferable knowledge across domains for cross-scene classification in RSIs under the PDA scenario. Specifically, a new bi-level reweighting strategy that incorporates transferability-aware patch selection and class-wise reweighting is developed to emphasize the transferable image patches and classes. Based on the aforementioned reweighting strategy, we further introduce a patch-weighted maximum mean discrepancy (PMMD) loss, which selectively aligns the distributions from the patch-level perspective, facilitating the learning of transferable domain-invariant representations. The experimental results of SAT demonstrate its effectiveness and superiority in addressing this practical domain adaptation (DA) task, outperforming state-of-the-art methods in PDA tasks on four datasets.**

***Index Terms*— Cross-scene classification, partial domain adaptation (PDA), remote sensing, vision transformer (ViT).**

I. Introduction

REMOTE sensing image (RSI) cross-scene classification has made tremendous progress in recent years, driven by the rapid advancement of deep learning approaches [1], [2], [3]. The most prevalent method to handle cross-scene classification tasks is domain adaptation (DA), which aims to apply knowledge obtained from a labeled source domain to a different unlabeled target domain [4], [5]. Existing DA methods usually assume an identical class space (closed-set DA) across domains while following different distributions [6], [7], [8], [9], [10]. These methods operate by either explicitly (metric-based) [6], [7], [8] or implicitly (adversarial-based) [9], [10] narrowing the gap between different distributions to learn domain-invariant feature representations across the domain. However, this assumption is frequently challenged in many real-world applications, as obtaining a priori knowledge about class information in unknown target tasks proves to be impractical [11], [12]. As the accessibility of remote sensing data from various satellite platforms continues to increase, a more practical and effective approach is to leverage the knowledge acquired from large-scale datasets with more diverse classes to address downstream tasks in the target datasets [13], [14]. In this scenario, the source domain includes a broader range of classes, while the classes in the target domain are a subset of those in the source domain, referred to as partial DA (PDA) [15], [16].

Compared to the previous standard closed-set DA methods, PDA methods show superior potential in real-world applications. The notable distinction between them lies in the presence of outlier classes within the source domain, which poses a significant challenge in addressing the domain shift problem. In this scenario, the transfer of knowledge from the entire domain may lead to negative transfer [17], [18]. This is because irrelevant knowledge from the outlier classes is also conveyed, leading to the misalignment of distributions. To mitigate the negative transfer problem in PDA, several successful attempts have focused on the reweighting of samples [15], [16], [19], [20]. Their key idea is to assign different weights by evaluating the transferable probability of data. Gradually increasing weights are assigned to transferable samples with high transferability, which are more likely to derive from shared class space. Consequently, the positive transfer is promoted, while the negative transfer of knowledge from the outlier classes is effectively suppressed.

In real-world applications of remote sensing, our focus is on identifying the labels of target entities among established candidate categories, rather than simply classifying them as unknown. Therefore, the setting of the PDA scenario becomes more practical and indispensable. Since many existing PDA methods have achieved promising performance, they also encounter significant challenges for cross-scene classification tasks in RSIs. As illustrated in Fig. 1, the variations in resolution, perspective, and imaging conditions in RSIs result in high interclass similarity and significant intraclass discrepancies among samples. Furthermore, RSIs commonly exhibit complex spatial backgrounds and local patterns, which may

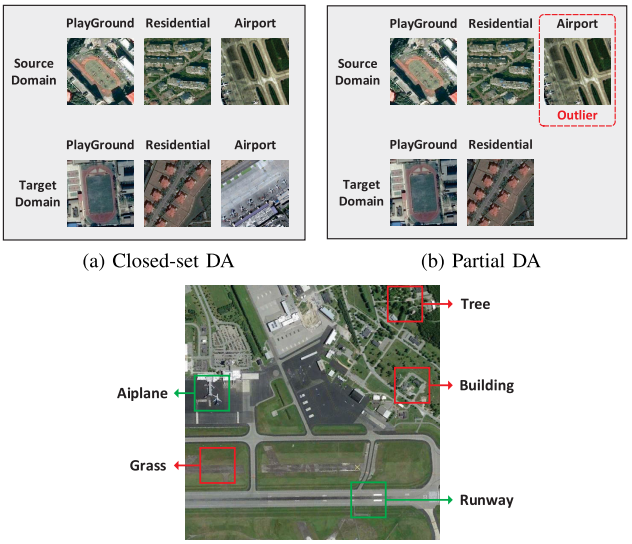
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(a) Closed-set DA (b) Partial DA

(c) Complex local patterns in RSIs

Fig. 1. Illustration of closed-set DA and PDA. (a) In the closed-set DA, the source and target domains share identical class space. (b) In the PDA, the target class space is considered to be a subset of the source domain. (c) In this scene, some distinct local patterns can be found, including "airplane," "runway," "building," "grassland," and "trees" within the airport.

be task-irrelevant or can also be found in samples from other categories, as shown in Fig. 1(c). Therefore, evaluating the transferability of semantic representations in RSIs becomes particularly challenging. Existing PDA methods may struggle to effectively handle the knowledge transfer of RSIs with various local patterns. Their allocation of equal attention to feature representations across all regions, without filtering out irrelevant knowledge, may result in the amplification of the negative transfer [21], [22]. Several studies have been proposed to investigate remote sensing applications under the PDA scenario, such as scene classification [23], [24] and crop yield prediction [25]. However, the previously mentioned challenges remain unresolved sufficiently.

Recently, the vision transformer (ViT) model [26] has provided a promising solution, leveraging its inherent advantage in processing regional patch-level representations. By the utilization of the self-attention mechanism [27], ViT treats images as sequences of patches, allowing greater flexibility in handling local patterns and global relationships within these patches. Several efforts have been made to employ transformers for addressing DA problem [28], [29], [30]. However, previous investigations have primarily focused on closed-set DA settings. In addition, relying solely on the ViT architecture as the backbone is insufficient. First, within the deep encoder layers, ViT's class token focuses more on global information and fails to filter out task-irrelevant local information. Second, ViT faces challenges similar to previous methods in effectively isolating source outlier classes and mitigating negative transfer within RSIs. The superiority of ViTs in the cross-scene classification tasks in RSIs is still to be explored, especially within the PDA scenario.

In this article, we propose a new selective alignment transformer (SAT) to effectively address the cross-scene classification challenges in RSIs under the PDA scenario. Specifically, we first introduce a bi-level reweighting strategy that includes class- and patch-level reweightings to emphasize the transferable knowledge across domains. Different from previous PDA methods that uniformly reweight the feature embeddings of entire samples, we tackle this problem from a patch-level perspective by further evaluating the transferability of image patches. This way, the model learns to focus more on the regions of the samples that are transferable. Meanwhile, class-level reweighting is also applied to constrain the feature alignment to the shared class space. Then, we further propose an improved patch-weighted maximum mean discrepancy (PMMD), which selectively aligns the distributions of image patches by considering both transferable patches and classes. By doing this, irrelevant knowledge from source outlier classes and complex local patterns in RSIs can be effectively filtered out, leading to improved cross-domain performance.

The primary contributions can be highlighted as follows.

1) A new SAT is proposed for RSIs cross-scene classification tasks in a more practical and challenging PDA scenario. This method is effective in addressing the task by selectively aligning transferable knowledge, thereby mitigating negative transfer.
2) A new bi-level reweighting strategy and a customized patch-weighted MMD (PMMD) loss are developed to fully explore transferable domain-invariant features while filtering out untransferable knowledge from the patch-level perspective.
3) The extensive experiments are conducted on four datasets for RSIs cross-scene classification tasks in the PDA scenario. Our proposed SAT outperforms other comparative methods, achieving state-of-the-art results and demonstrating the superiority of the method.

In Section II, we review the relevant research related to our method. In Section III, we present a detailed description of SAT. In Section IV, the experimental setups and datasets are outlined. Then, comprehensive experiments are conducted to thoroughly assess the effectiveness of our method. Finally, we conclude this article.

II. Related Work

We conduct a review of the relevant research in this article, mainly including partial DA methods, ViT-based DA methods, and DA for cross-scene classification in RSIs.

A. PDA for Classification

The objective of DA is to employ labeled data from the source domain to train a model that can generalize well to diverse target domain datasets. Closed-set DA, as a standard setting of DA methods, is mainly employed to solve distribution shift issues across domains with identical class space [6], [7], [8], [9], [10]. These methods can be categorized into two groups. The first group of DA methods matches the distributions by explicitly reducing a statistical metric of discrepancy across domains, such as maximum mean discrepancy (MMD) [7], and CORAL [8]. The second involves adversarial learning-based DA methods, employing an adversarial training paradigm within generative adversarial networks (GANs) [31]

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to narrow the distribution discrepancies across domains. The main idea involves the learning of domain-invariant feature representations by engaging in an adversarial game between the feature encoder and the discriminator [9], [10], [32].

Different from closed-set DA, PDA addresses a practical scenario where the target domain contains only a subset of the class space present in the source domain. Moreover, PDA poses an additional challenge of aligning distributions across domains while effectively handling the transfer of irrelevant knowledge caused by source outlier classes. Most PDA methods follow the paradigm of sample reweighting, reassigning different weights to source domain samples based on the transferability of the data. Samples from shared classes are characterized by a high transferable probability and receive increased weights during the training process. Conversely, samples with low transferability are considered to be taken from outlier classes and are given reduced weights. Typically, PADA [15] evaluates class-level transferability by computing the average label predictions of the target domain samples. In [16], ETN introduces a special classifier to reweight the samples during the training process, which can effectively tackle the limitations of the class-level weighting methods in the PDA scenario. Zhang et al. [19] reweight samples in feature alignment via transferability information from an additional domain discriminator. In [20], an adversarial reweighting approach is proposed to learn the weights of data. The BA^{3}US proposes a balanced adversarial alignment technique for adjusting the weights of both the classifier and the domain discriminator [33]. In [34], a combined reweighting strategy, along with a contrastive learning-based distribution alignment strategy, is proposed to mitigate negative transfer in PDA. DRCN [35] employs a residual correction block to facilitate the transferring of task-level knowledge. Kim and Hong [36] propose the AGAN to reduce negative transfer by exploring intradomain and interdomain information. In [37], RTNet considers the multilevel information to address the PDA tasks with the help of reinforcement learning.

We argue that the existing sample-level reweighting mechanism is still insufficient in addressing the negative transfer in RSIs. Unlike previous PDA methods, which solely utilize feature representations of entire samples, we solve this problem by exploring the transferability of samples from both patch- and class-level perspectives during distribution alignment, thereby facilitating selective knowledge transfer.

B. ViT in DA

With the significant progress made by transformers in natural language processing (NLP) tasks, some studies have started applying transformer architecture to visual tasks, resulting in state-of-the-art performance. In [26], ViT is proposed to tackle image classification tasks leveraging the self-attention mechanism. Unlike conventional convolutional neural networks (CNNs), ViT can generate robust feature representations by capturing long-range relations among image patches across the entire image. Thanks to this, ViT has demonstrated promising performance across multiple visual tasks, such as image classification [26], object detection [38], and semantic segmentation [39]. Recently, researchers have been exploring the potential of transformers for cross-domain applications. Xu et al. [28] propose CDTrans that integrates both self- and cross-attention mechanisms for distribution alignment. In [29], TAM is injected into the transformers, aiming to highlight transferable and discriminative features. Sun et al. [30] propose SSRT to achieve a safe distribution alignment through the utilization of a feature refinement strategy. Beyond the closed-set DA scenario, there has been limited exploration of transformers in the PDA setting. Our method is specifically designed for this practical application in RSIs, making it more effective in addressing the challenges.

C. DA in RSI

With the notable advancements in DA techniques, numerous DA-based methods for cross-scene classification of RSIs have emerged, which aims to narrow the distribution gaps between domains resulting from different imaging platforms. In [40], CaGAN explicitly models the class-level discrepancies across domains for semantic segmentation tasks in RSIs. In [41], a two-branch DA framework is proposed to tackle cross-scene classification tasks of hyperspectral images (HSIs) by separately handling spatial and spectral information. Huang et al. [42] propose a weighted DA network to filter noisy samples for cross-domain wetland mapping tasks. In [43], a task-specific classifier is employed to leverage the classification information effectively. In [44], AMRAN addresses the multiscale issue in cross-scene classification tasks to achieve more comprehensive distribution alignment. Zhang et al. [45] propose the DFENet to further enhance the discriminativeness of feature representations. Niu et al. [46] propose MRDAN to acquire enhanced domain-invariant features through the proposed dynamical distribution alignment approach. Zhao et al. [47] introduce a new DA framework by exploiting multilevel features and decision boundaries across domains. Ling et al. [48] propose a multibranch DA method incorporating a cross-attention mechanism to address the challenge of small-sized targets in the cross-scene classification of HSIs.

The previously mentioned methods are mainly focusing on the standard closed-set DA scenario. In addition, several studies have explored cross-scene classification tasks for RSIs in open-set DA scenario [49], [50] and universal DA [51] scenario. In open-set DA tasks, outlier classes appear in the target domain. In this framework, target-only classes are identified solely as unknown categories without any further classification. The universal DA scenario occurs when the outlier classes are present in both domains simultaneously. The scenario of PDA is highly relevant in real-world applications, as it is practical and feasible to adapt models from a broader source domain to a smaller domain. In [23], CPADA is proposed to transfer relevant samples while ignoring irrelevant ones by employing a coordinate loss. Zheng et al. [24] propose a novel learning strategy to strengthen predictions for transferable data while mitigating the potential of negative transfer. PDANN [25] is proposed to address crop yield prediction tasks by reducing the importance of outlier samples. Thus,

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we propose SAT for the cross-scene classification of RSIs, aiming to facilitate the application of PDA methods in real-world scenarios.

III. Selective Alignment Transformer

A new SAT is proposed in this article for the cross-scene classification of RSIs under the PDA scenario. Our method aims to simultaneously explore the transferable knowledge of PDA from both patch- and class-level perspectives to improve cross-domain performance. This selective alignment of transferable patches can efficiently filter out irrelevant knowledge, achieving selective feature alignment.

Specifically, we first propose a bi-level reweighting strategy, which simultaneously highlights the transferable image patches and classes, facilitating the exploration of domain-invariant knowledge while restricting the knowledge transfer within the shared class space. Second, our method incorporates a patch-weighted MMD (PMMD) to guide the feature adaptation toward the most transferable regions, thereby facilitating enhanced positive transfer. In summary, our SAT can mitigate negative transfer for effective distribution alignment. Fig. 2 illustrates the comprehensive structure of SAT.

A. Problem Formulation

Our aim is to improve the cross-domain performance on target domain datasets in the PDA scenario by leveraging the knowledge from the labeled source domain. In the paradigm of DA, the labeled source domain dataset $\mathcal{D}_s = \{(x_i, y_i)\}_{i=1}^{n_s}$ with n_s samples and the unlabeled target domain dataset $\mathcal{D}_t = \{(x_i)\}_{i=1}^{n_t}$ with n_t samples are provided. These two datasets are drawn from different distributions p and q, i.e., $p \neq q$. In addition, the class spaces of the source domain and the target domain are denoted as $\mathcal{C}_s$ and $\mathcal{C}_t$, respectively. Different from in standard closed-set DA ($\mathcal{C}_s = \mathcal{C}_t$), the class space $\mathcal{C}_t$ is a subset of $\mathcal{C}_s$ in PDA scenario, i.e., $\mathcal{C}_t \subset \mathcal{C}_s$. We represent the outlier class space in the source domain that is $\bar{\mathcal{C}}_s = \mathcal{C}_s \backslash \mathcal{C}_t$. The shared class space between them is $\mathcal{C}$ such that $\mathcal{C} = \mathcal{C}_t$.

B. ViT Framework

In SAT, we augment the ViT by incorporating a domain token $\mathbf{z}_0^{N+1} = \mathbf{x}_{\text{domain}} \in \mathbb{R}^D$, accompanied by an auxiliary domain discriminator D_0 for the original patch embeddings. The domain embedding allows our model to quantify the transferability of samples from a patch-level perspective, facilitating selective distribution alignment for PDA tasks. In the vanilla ViT, the image $\mathbf{I} \in \mathbb{R}^{C \times H \times W}$ is split into N fixed-size flattened patches $\mathbf{x} \in \mathbb{R}^{N \times (P^2 \cdot C)}$, where H, W, and C represent the height, width, and channels of the original image, (P, P) is the resolution of each image patch, and we have $N = HW/P^2$. The patch embeddings are obtained by mapping these flattened patches into a sequence of D-dimensional vectors with a trainable encoder $\mathbf{E} \in \mathbb{R}^{(P^2 \cdot C) \times D}$. A trainable classification token $\mathbf{z}_0^0 = \mathbf{x}_{\text{class}} \in \mathbb{R}^D$ is included in the sequence for classification tasks, which aims to extract the global representations. The domain token is utilized similar to the class token; both of them are integrated into the transformer through the attention mechanism. The distinction lies in its objective, which is to predict domain labels rather than class labels of the data. Finally, we obtain the input sequence of embedded vectors $\mathbf{z}_0^0$ for the SAT model

$$\mathbf{z}_0 = \left[\mathbf{x}_{\text{class}}; \mathbf{x}_p^1\mathbf{E}; \mathbf{x}_p^2\mathbf{E}, \ldots, \mathbf{x}_p^N\mathbf{E}; \mathbf{x}_{\text{domain}}\right] + \mathbf{E}_{\text{pos}} \tag{1}$$

where the embeddings $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N+2) \times D}$ represent learnable position embeddings that are added to the patch embeddings.

After patch embedding, the embedded vectors are input into the transformer encoder to produce the encoded embeddings. Within the transformer encoder, there are L layers composed of multihead self-attention (MSA) and multilayer perceptron (MLP) blocks. The self-attention mechanism models and processes sequences by weighting elements across various positions in the input sequence. The self-attention mechanism comprises three elements: queries ($\mathbf{Q} \in \mathbb{R}^{N \times D}$), keys ($\mathbf{K} \in \mathbb{R}^{N \times D}$), and values ($\mathbf{V} \in \mathbb{R}^{N \times D}$). By calculating the correlation between the input queries and keys, and then using these correlation weights to weight the value vectors, the output is ultimately generated. This mechanism allows the model to effectively capture the global semantic dependencies within images. The MSA process can be formulated as follows:

$$\text{MSA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V}. \tag{2}$$

Thus, we can obtain the output of the ViT model as follows:

$$\begin{aligned} 2\mathbf{z}'_\ell &= \text{MSA}\left(\text{LN}\left(\mathbf{z}'_{\ell-1}\right)\right) + \mathbf{z}_{\ell-1}, \quad \ell = 1, \ldots, L \\ \mathbf{z}_\ell &= \text{MSA}\left(\text{LN}\left(\mathbf{z}'_\ell\right)\right) + \mathbf{z}'_\ell, \quad \ell = 1, \ldots, L \\ y_{\text{class}} &= \text{LN}\left(\mathbf{z}_L^0\right), \\ y_{\text{domain}} &= \text{LN}\left(\mathbf{z}_L^{N+1}\right) \end{aligned} \tag{3}$$

where $\text{LN}(\cdot)$ is the layer normalization and the output of ℓth transformer encoder layer $\mathbf{z}_\ell$ can be represented as follows:

$$\mathbf{z}_\ell = \left[\mathbf{x}_{\text{class}}; \mathbf{z}_\ell^1, \mathbf{z}_\ell^2, \ldots, \mathbf{z}_\ell^N; \mathbf{x}_{\text{domain}}\right] \tag{4}$$

where $\mathbf{z}_l^i$ is the ith encoded embeddings of image patches. Finally, the learned class token $\mathbf{z}_L^0$ for classification and the domain token $\mathbf{z}_L^{N+1}$ for predictions of domain labels are fed into an MLP followed by a softmax classifier to obtain the classification results.

C. Bi-Level Reweighting Strategy

Reweighting samples via the transferability information has been proven to be an efficient strategy to solve PDA problem [15], [16], [18], [19], [20]. The previous reweighting strategy in PDA typically involves assigning weights to instances or entire features that exhibit great relevance to the target domain. However, this method may introduce noisy information, as the entire sample could aggregate much information from noisy patches. In this scenario, ineffective filtering of irrelevant knowledge from outlier classes $\bar{\mathcal{C}}_s$ leads to poor distribution alignment. To address this, we employ a bi-level reweighting strategy, which simultaneously assigns weights to each class and patch to highlight the transferable domain-invariant knowledge for selective distribution alignment. It mainly includes two parts: transferability-aware patch selection and class-wise reweighting.

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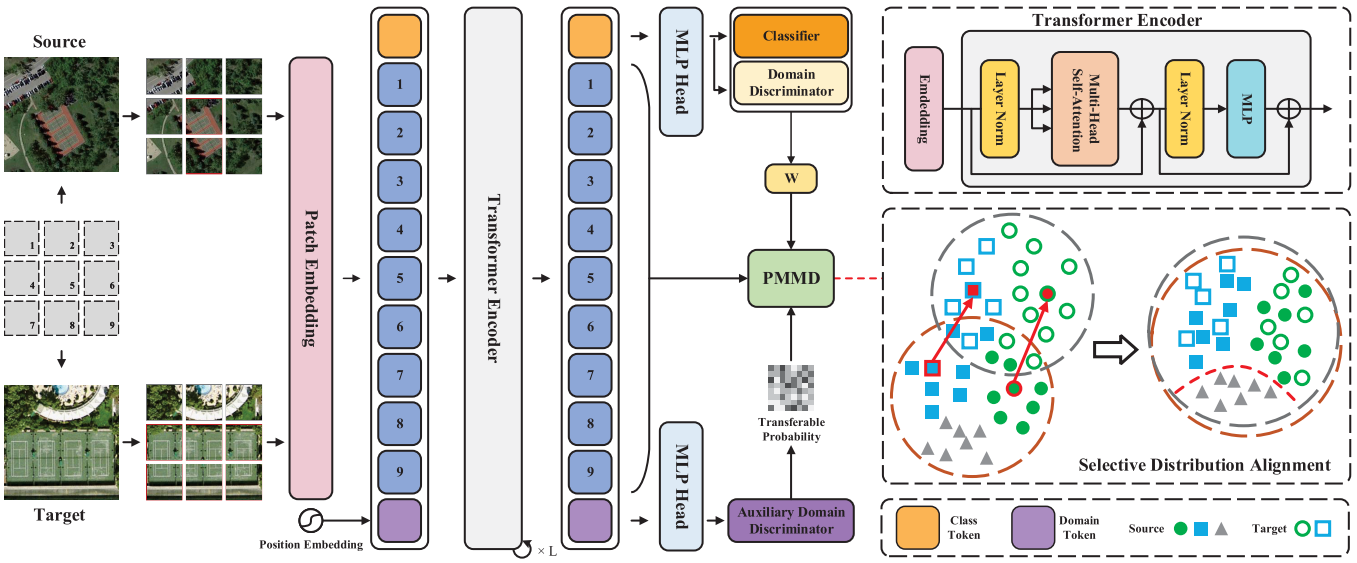


Fig. 2. Illustration of our SAT, where a domain token in the transformer layer and an auxiliary domain discriminator are developed to effectively evaluate the transferability of image patches. A customized PMMD loss is also proposed, which selectively aligns sample patches with transferability. This way, the domains will be selectively aligned while isolating the source outlier classes and mitigating negative transfer.

1) Transferability-Aware Patch Selection: The purpose of this component is to further evaluate the transferability of image patches. In the prior work [16], the predicted probability $\hat{y} = D_0(\mathbf{z}^i)$ over the source class space $\mathcal{C}_s$ is utilized to quantify the transferability of data. Each sample is reweighted by an entropy-based weight $w(\mathbf{x}_i) = 1 + e^{-H(\hat{y}_i)}$, where $H(\hat{y}_i) = -\hat{y}_i \log(\hat{y}_i)$ is the conditional-entropy and the $\hat{y}_i$ is the label prediction for sample $\mathbf{x}_i$. A large value of $w(\mathbf{x}_i)$ indicates that the sample exhibits increased similarity to the target domain, suggesting a high probability of its presence in the shared class space $\mathcal{C}$. Different from previous DA methods, SAT employs a sophisticated evaluation of transferability for image patches.

Given the encoded class token $\mathbf{z}_L^0$ of the last transformer layer, we can obtain the attentive weights $\mathbf{A}_\ell \in \mathbb{R}^{(N+2)\times D}$ in ℓth layer for N image patches by calculating the correlation between the input queries and keys

$$\mathbf{A}_\ell = \left[a_\ell^1, a_\ell^2, \ldots, a_\ell^{N-1}, a_\ell^N\right]. \tag{5}$$

For the MSA, we initially compute the average of attention scores across all heads.

An intuitive idea is to select tokens with significant attentive weights associated with class tokens generated through the MSA mechanism. However, the optimal task-specific embeddings may fail to effectively capture transferable features that can be generalized to other datasets. To this end, we employ a mutual attention weighting strategy, which enhances a more balanced and comprehensive focus on transferable patch-level details by integrating patch transferability and task-specificity.

The domain attentive weights $\mathbf{D}_\ell \in \mathbb{R}^{(N+2)\times D}$, which are queried by the domain token $\mathbf{z}_\ell^{N+1}$ for image patches, can be denoted as follows:

$$\mathbf{D}_\ell = \left[d_\ell^1, d_\ell^2, \ldots, d_\ell^{N-1}, d_\ell^N\right]. \tag{6}$$

The final patchwise weights w_p within sample $\mathbf{x}_i$ obtained by our mutual attention weight strategy can be calculated as follows:

$$w_p\left(\mathbf{z}^j\right) = a_L'^j \cdot d_L'^j \cdot w(\mathbf{x}_i) \tag{7}$$

$$a'^j_L = \frac{e^{a_L^j}}{\sum_{i=1}^N e^{a_L^i}}, \quad d'^j_L = \frac{e^{d_L^j}}{\sum_{i=1}^N e^{d_L^i}}. \tag{8}$$

To mitigate the potential separation of task-specific and transferable knowledge when the attentive weights queried by these tokens differ significantly, we further employ a consistency constraint loss on the transformer layer as follows:

$$\mathcal{L}_c = \sum_{a_l^i < a_l^j} e^{a_l^j} f\left(d_l^j - d_l^i\right) \tag{9}$$

where $e^{d_l^j}$ is the weighted term of hinge loss $f(x) = \max\{x, 0\}$. This consistency constraint loss promotes $d_l^j > d_l^i$ when $a_l^i > a_l^j$. It encourages the crucial patches (i.e., patches with large attentive weights queried by class token) that also exhibit high transferability. This can effectively highlight embeddings that are both discriminative and transferable across domains.

2) Class-Wise Reweighting: Uniformly assigning the same weight to various classes may result in a negative transfer. To guarantee effective knowledge transfer, we employ class-wise reweighting to reduce the classifier's emphasis on outlier classes. For the target domain data, the model is more likely to predict them as belonging to shared classes rather than outlier classes. Hence, we can use the final predictions on the target domain to adjust the weighting of the classes [18]. The class-wise weights $\mathbf{w}_c$ for the class k can be calculated as follows:

$$w_c^k = \sum_{i=1}^{N_t} C\left(E\left(\mathbf{x}_{t,k}^i\right)\right)/n_t \tag{10}$$

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where w_c^k is the kth entry of vector $\mathbf{w}_c$. The weights are further normalized as $\mathbf{w}_c = \mathbf{w}_c / \max(w_c^k)$.

D. PDA With Patch-Weighted Maximum Mean Discrepancy

Considering the varying importance of local patches in distribution alignment, we further utilize the reweighting weights based on the transferability of image patches to selectively narrow the distribution discrepancies among them. While the MSA mechanism in the transformer effectively captures long-range dependencies between local image patches, the class token in the deep encoder layers focuses more on global information. This potentially limits the model's ability to effectively evaluate the attention of regions carried by image patches for distribution alignment. Global alignment can harm the cross-scene performance due to the lack of the selection of crucial local features, essential for applications with complex local patterns. Therefore, selective alignment of distributions is essential to mitigate negative transfer resulting from global alignment.

As an estimation of dissimilarity between two distributions, MMD has been extensively employed in various metric-based DA methods by explicitly minimizing this measure [6], [7], [8], [52]. The MMD loss can be represented as follows:

$$\mathcal{L}_{\mathrm{MMD}} = \left\| \frac{1}{n_s} \sum_{\mathbf{x}_i \in \mathcal{D}_s} \phi(\mathbf{x}_i) - \frac{1}{n_t} \sum_{\mathbf{x}_j \in \mathcal{D}_t} \phi\left(\mathbf{x}_j\right) \right\|_{\mathcal{H}}^2 \tag{11}$$

where $\phi(\cdot)$ is a projection function to map the samples reproducing kernel Hilbert space (RKHS).

Building on the foundation of the previous bi-level reweighting framework, we further propose the PMMD to align the distributions from the patch-level perspective

$$\mathcal{L}_{\mathrm{PMMD}} = \frac{1}{k} \sum_{k=1}^{\mathcal{C}} \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \sum_{j=1}^{N} \mu_{s,c}^k w_{s,p}^j \phi\left(\mathbf{z}_s^{i,j}\right) - \frac{1}{n_t} \sum_{i=1}^{n_t} \sum_{j=1}^{N} \mu_{t,c}^k w_{t,p}^{i,j} \phi\left(\mathbf{z}_t^{i,j}\right) \right\|_H^2 \tag{12}$$

where $\mathbf{z}_c^{i,j}$ is the output derived from the jth encoded patch of samples $\mathbf{x}^i$ in the last transformer encoder layer and μ_c^i and w_p^j represent the class weights for samples within class k and the weights for image patches, respectively. The indices "s" and "t" represent samples originating from the source or target domains. The weight μ_c^i for samples within class k can be obtained as follows:

$$\mu^k = \frac{w_c^k y_k^i}{\sum_{(x_j, y_j) \in D} y_k^j}. \tag{13}$$

The prediction probability $\hat{y}_t^i$ is employed for the date from the target domain, replacing the use of true labels y^i. Our PMMD differs from the previous MMD as it considers the task-specific and transferable information at the patch-level of each sample by simultaneously assigning weights to patches and classes across all images. This adjustment is well-suited for PDA tasks in remote sensing applications, helping to isolate outlier classes and filtering irrelevant knowledge in complex scenarios.

E. Global Distribution Alignment

We adopt the adversarial DA, which employs a max–min game between $E(\cdot)$ and $D(\cdot)$, encouraging the network to learn domain-invariant feature representations [9], [10]. In addition, we propose to minimize the entropy of class probabilities on the target domain to facilitate discriminative feature learning.

1) Adversarial DA: Following [16], we align the distributions by weighting source samples according to the transferable probabilities. This allows the model to focus on aligning the class-conditional distributions in the shared class space $\mathcal{C}$, thereby mitigating negative transfer.

The adversarial adaptation loss for global distribution alignment is represented as follows:

$$\mathcal{L}_D = -\frac{1}{n_s} \sum_{k=1}^{c} \sum_{i=1}^{n_s} w\left(\mathbf{x}_{s,k}^i\right) d_i \log\left(D\left(\mathbf{z}_{i,s}\right)\right) - \frac{1}{n_t} \sum_{i=n_s+1}^{n_s+n_t} w\left(\mathbf{x}_{t,k}^i\right)(1-d_i) \log\left(1 - D\left(\mathbf{z}_{i,t}\right)\right). \tag{14}$$

The auxiliary domain discriminator D_0 is trained similar to discriminator D with the cross-entropy loss. However, it focuses on domain tokens instead of class tokens. Moreover, D_0 is not included in adversarial training.

2) Source Supervised Training: The cross-entropy loss is also employed to ensure the discriminativeness of the feature embeddings. Specifically, the transferable class-wise weights $w(\mathbf{x}_i)$ of samples are involved into the training of the classifier C

$$\mathcal{L}_{\mathrm{cls}} = -\frac{1}{n_s} \sum_{k=1}^{c} \sum_{i=1}^{n_s} w_c^k y_i^k \log\left(C\left(\mathbf{z}_{i,s}\right)\right) \tag{15}$$

where y_i is the true label of samples in the source domain and w_k is the kth entry of w, indicating the probability of a sample belonging to the shared class space $\mathcal{C}$.

3) Target Discriminative Clustering: Similar to [6] and [53], SAT leverages the entropy minimization principle in the predicted class probabilities to facilitate effective knowledge transfer. This can be treated as a self-training regularization on the target domain, which helps reduce uncertainty or confusion in predictions, thereby enhancing the generalization ability and discriminativeness of learned feature embeddings on the target data

$$\mathcal{L}_{\mathrm{self}} = \frac{1}{n_t} \sum_{i=1}^{n_t} \hat{y}_i^t \log\left(\hat{y}_i^t\right). \tag{16}$$

F. Overall Objective and Training

The comprehensive objective of the SAT can be summarized as

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{cls}} + \mathcal{L}_{\mathrm{adv}} + \lambda \mathcal{L}_{\mathrm{PMMD}} + \varphi \mathcal{L}_{\mathrm{self}} + \gamma \mathcal{L}_c \tag{17}$$

where λ, φ, and γ are the tradeoff weights for each loss. The gradient reversal layer (GRL) layer is utilized to invert the gradient for feature encoder [10]. In addition, we have $\mathcal{L}_{\mathrm{adv}} = \mathcal{L}_D + \mathcal{L}_{D_0}$ in (13). As the auxiliary domain discriminator D_0 is

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trained similar to the discriminator D, the SAT can be trained by a mini–max optimization

$$\left(\hat{\theta}_E, \hat{\theta}_C, \hat{\theta}_{D_0}\right) = \arg\min_{\theta_F, \theta_C, \theta_{D_0}} \mathcal{L}_{\text{total}}\left(\theta_E, \theta_C, \hat{\theta}_D, \theta_{D_0}\right)$$
$$\left(\hat{\theta}_D\right) = \arg\max_{\theta_D} \mathcal{L}_{\text{total}}\left(\hat{\theta}_E, \hat{\theta}_C, \theta_D, \hat{\theta}_{D_0}\right) \quad (18)$$

where θ_E, θ_C, θ_D, and θ_{D_0} are the parameters of transformer encoder, classifier, domain discriminator, and auxiliary domain discriminator.

IV. Experiments and Analysis

A. Datasets

We utilize four publicly available datasets to create cross-scene classification datasets and validate the cross-scene performance.

1) AID [54]: The AID dataset comprises 30 aerial scene categories and includes a total of 10 000 images taken under diverse regional and imaging conditions. It serves as a standardized benchmark for assessing the accuracy and effectiveness of classification methods in the context of aerial imagery. The range of image resolutions extends from 8 to 0.5 m, maintaining a fixed size of 600 × 600 pixels.

2) NWPU-RESISC45 [55]: The dataset comprises 31 500 RSIs distributed among 45 scene categories. Each category consists of 700 RSIs sized at 256 × 256 pixels with a spatial resolution ranging from 30 to 0.2 m, acquired from Google Earth.

3) UC Merced [56]: This dataset includes images that represent 21 diverse scene categories, and each category comprises precisely 100 images, each sized at 256 × 256 pixels.

4) PatternNet [57]: The PatternNet dataset consists of 38 categories, each with 800 RSIs at a resolution of 256 × 256 pixels. The dataset is acquired from Google Maps for the purpose of RSI retrieval and classification.

Based on these datasets, 11 cross-scene classification tasks are established within the scenario of PDA. These tasks include A ↔ N, A ↔ P, A ↔ U, N ↔ P, N → U, and P ↔ U. The detailed category information of these tasks is presented in Table I. For each task, we retain the private classes in the source domain dataset while removing the corresponding samples from the private classes in the target domain. It should be noted that some categories sharing identical meanings but employing different names (noted by parentheses) have been merged. In the U → N task, all source classes are incorporated into the target domain, thus making this task a standard closed-set DA. In Fig. 3, some sample images from the aforementioned datasets are displayed. The differences in sensors, resolutions, spectral ranges, and other factors could result in considerable distribution discrepancies across images collected from different satellite platforms. Furthermore, we can observe that RSIs acquired from identical satellite platforms also tend to exhibit significant intraclass diversity and interclass similarity.

B. Implementation Details

The experiments are conducted using the PyTorch framework on the workstation equipped with four NVIDIA GeForce RTX 4090 GPUs. All images from datasets are resized to the dimension of 224 × 224, serving as input for the model. The pretrained weights on ImageNet are utilized in our experiments. We fine-tune all models on each task using the stochastic gradient descent (SGD) optimizer, with an initial learning rate of $\eta_0 = 3 \times 10^{-3}$ and a momentum of $\beta = 0.9$. The learning rate progressively decreases during training according to the following strategy: $\eta_p = \eta_0/(1+\alpha p)^{\varepsilon}$, where p denotes the training progress, linearly transitioning from 0 to 1. The parameters α and ε are set to 10 and 0.75, respectively. The tradeoff weights λ, φ, and γ in the overall loss function $\mathcal{L}_{\text{total}}$ are fixed to 1, 0.09, and 0.1 in our experiments. The parameter δ in the GRL, which controls the power of adversarial training, is progressively increased from 0 to 1. Finally, due to the network architecture and performance, the total training steps for each task are established as 10 000 with a batch size of 32 for both domains.



Fig. 3. Illustration of some examples from four datasets.

C. Experimental Results

We demonstrate the effectiveness of our SAT by comparing it with state-of-the-art methods, including CNN-based partial DA methods, IWAN [19], ETN [16], SPDA [58], BA^{3}US [33], AR [20], and I-DANN [24], and Transformer-based DA methods, CDTrans [28], TVT [29], and SSRT [30]. We also include source-only models utilizing ResNet50 and ViT backbones for comparison. The "-S/B" indicates that the experimental results are obtained based on ViT small/base backbones, varying in width and number of heads in MSA. We adopt ViT-S/B as the backbones of our method and ViT-based DA methods. All CNN-based DA methods use ResNet50 as their backbones.

1) Experimental Results on Closed-Set DA Scenario: For a comprehensive evaluation of our method, we conducted comparative experiments on closed-set DA tasks. The experimental results for the four datasets are presented in Table II. This proves that even in a closed-set setting, our method can still effectively reduce distribution discrepancies. It is noticeable that the source-only method with ViT-B can achieve an average accuracy of 80.71%, which outperforms the CNN-based source-only model and DANN. This is mainly due

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TABLE I
DATASETS AND CATEGORIES DETAILS FOR PDA TASKS

Task	Private Classes of Right Dataset	Private Classes of Left Dataset	Shared Classes
A ↔ N	BareLand, Center Park, Pond, Resort, School, Square	Airplane, Basketball Court, Chaparral, Cloud, Freeway, Golf Course, Intersection, Island, Lake, Mobile Home Park, Palace, Railway, Roundabout, Runway, Sea Ice, Ship, Snowberg, Tennis Court, Terrace, Thermal Pow Station, Wetland	Airport, Baseball Field (Baseball Diamond), Beach, Bridge, Church, Commercial (Commercial Area), Dense Residential, Desert, Farmland (Circular Farmland, Rectangular Farmland), Forest, Industrial (Industrial Area), Meadow, Medium Residential, Mountain, Parking (Parking Lot), Playground (Ground Track Field), Port (Harbor), Sparse Residential, Railway Station, River, Stadium, Storage Tanks, Viaduct (Overpass)
A ↔ P	Airport, BareLand, Center, Church, Commercial, Desert, Farmland, Industrial, Meadow, Medium-Residential, Mountain, Park, Pond, RailwayStation, Resort, School, Square, Stadium	Airplane, Basketball Court, Cemetery, Chaparral, Christmas Tree Farm, Closed Road, Coastal Mansion, Crosswalk, Ferry Terminal, Freeway, Golf Course, Intersection, Mobile Home Park, Nursing Home, Oil Gas Field, Oil Well, Parking Space, Railway, Runway, Runway Marking, Shipping Yard, Solar Panel, Swimming Pool, Tennis Court, Transformer Station, Wastewater Treatment Plant	Baseball Field, Beach, Bridge, Forest, Dense Residential, Parking (Parking Lot), Port (Harbor), River, Sparse Residential, Storage Tanks, Viaduct (Overpass), Playground (Football Field)
A ↔ U	Airport, Bare Land, Bridge, Center, Church, Desert, Industrial, Meadow, Mountain, Park, Playground, Pond, Railway Station, Resort, School, Square, Stadium	Airplane, Chaparral, Freeway, Golf Course, Intersection, Mobile Home Park, Runway, Tennis Court	Baseball Field (Baseball Diamond), Beach, Commercial (Buildings), Dense Residential, Farmland (Agricultural), Forest, Medium Residential, Parking (Parking Lot), Port (Harbor), River, Sparse Residential, Storage Tanks, Viaduct (Overpass)
N ↔ P	Airport, Church, Circular Farmland, Cloud, Commercial Area, Desert, Industrial Area, Island, Lake, Meadow, Medium Residential, Mountain, Palace, Railway Station, Rectangular Farmland, Roundabout, Sea Ice, Ship, Snowberg, Stadium, Terrace, Thermal Power Station, Wetland	Cemetery, Christmas Tree Farm, Closed Road, Coastal Mansion, Crosswalk, Ferry Terminal, Nursing Home, Oil Gas Field, Oil Well, Parking Space, Runway Marking, Shipping Yard, Solar Panel, Swimming Pool, Transformer Station, Wastewater Treatment Plant	Airplane, Baseball Diamond (Baseball Field), Basketball Court, Beach, Bridge, Chaparral, Dense Residential, Freeway, Forest, Golf Course, Ground Track Field (Football Field), Harbor, Intersection, Mobile Home Park, Overpass, Parking Lot, Railway, River, Runway, Sparse Residential, Storage Tank, Tennis Court
N ↔ U	Airport, Basketball Court, Bridge, Church, Cloud, Desert, Ground Track Field, Industrial Area, Island, Lake, Meadow, Mountain, Palace, Railway, Railway Station, Roundabout, Sea Ice, Ship, Snowberg, Stadium, Terrace, Thermal Power Station, Wetland	None	Airplane, Baseball Diamond, Beach, Chaparral, Commercial Area (Buildings), Dense Residential, Forest, Freeway, Golf Course, Harbor, Intersection, Circular Farmland and Rectangular Farmland (Agricultural), Mobile Home Park, Medium Residential, Parking Lot, Overpass, River, Runway, Sparse Residential, Storage Tank, Tennis Court
P ↔ U	Basketball Court, Bridge, Cemetery, Christmas Tree Farm, Closed Road, Coastal Mansion, Crosswalk, Ferry Terminal, Football Field, Nursing Home, Oil Gas Field, Oil Well, Parking Space, Railway, Runway Marking, Shipping Yard, Solar Panel, Swimming Pool, Transformer Station, Wastewater Treatment Plant	Agricultural, Buildings, Medium Residential	Airplane, Baseball Field (Baseball Diamond), Beach, Chaparral, Dense Residential, Forest, Freeway, Golf Course, Harbor, Intersection, Mobile Home Park, Overpass, Parking Lot, Runway, River, Sparse Residential, Storage Tank, Tennis Court

TABLE II
CLASSIFICATION ACCURACY (%) OF CLOSED-SET DA CROSS-SCENE CLASSIFICATION TASKS IN RSIS

Methods	Backbone	Datasets												
		A → N	A → P	A → U	N → A	N → P	N → U	P → A	P → N	P → U	U → A	U → N	U → P	Average
Source-Only	ViT-B/16	79.99	90.79	75.03	92.93	90.54	88.81	83.16	63.41	80.89	66.44	65.15	91.41	80.71
SSRT-B	ViT-B/16	90.25	**99.82**	93.77	95.49	**98.83**	94.76	**98.74**	87.84	90.28	96.59	87.29	99.18	94.40
CDTrans	ViT-B/16	88.78	95.64	88.63	94.51	93.48	93.85	94.47	84.31	**94.89**	92.42	81.98	96.45	91.62
SAT-B	ViT-B/16	**92.61**	99.73	**97.62**	**97.70**	98.34	**97.10**	98.71	**88.03**	94.67	**96.90**	**88.19**	**99.61**	**95.77**

to the powerful capability of the transformer to capture long-range dependencies, leading to more versatile feature representations. Meanwhile, our method demonstrates competitive performance with advanced DA methods such as SSRT and CDTrans under standard closed-set DA setting, achieving an average accuracy of 95.77%. This is because the tasks with the PDA setting can be regarded as a special case of standard closed-set DA. Evaluating the transferability of samples and categories can also facilitate positive transfer in closed-set DA tasks.

2) Experimental Results on PDA Scenario: As for PDA tasks, we compare our method with state-of-the-art methods

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TABLE III
CLASSIFICATION ACCURACY (%) OF PDA CROSS-SCENE CLASSIFICATION TASKS IN RSIS

Methods	Backbone	Datasets											
		A → N	A → P	A → U	N → A	N → P	N → U	P → A	P → N	P → U	U → A	U → P	Average
Source-Only	ResNet50	70.39	69.04	57.15	81.68	85.51	69.48	51.05	44.61	55.22	53.19	91.17	66.23
IWAN	ResNet50	75.08	82.47	60.00	81.58	86.51	79.29	71.39	64.84	71.89	66.50	94.65	75.84
ETN	ResNet50	76.79	72.35	58.09	86.17	84.38	76.33	65.91	59.45	68.81	70.97	94.23	73.95
SPDA	ResNet50	85.14	74.08	62.72	84.77	90.92	81.43	69.70	59.25	80.56	70.42	**97.83**	77.89
BA^{3}US	ResNet50	79.32	80.12	57.08	87.88	89.97	83.38	67.83	65.75	75.39	69.82	97.67	77.66
AR	ResNet50	81.59	76.40	65.46	88.85	87.47	80.29	64.99	67.60	75.44	79.12	93.58	78.25
I-DANN	ResNet50	84.61	-	64.62	89.54	-	80.72	-	-	-	79.71	-	-
Source-Only	ViT-B/16	64.84	64.43	63.77	83.84	90.90	76.43	50.61	45.28	63.44	66.45	89.83	69.07
SSRT-S	ViT-S/16	80.02	74.23	63.54	82.24	88.94	77.86	50.33	64.35	67.33	72.75	93.30	74.08
SSRT-B	ViT-B/16	80.51	84.50	72.08	85.90	92.91	80.71	75.96	78.42	79.89	84.02	95.19	82.74
TVT	ViT-B/16	79.88	78.55	70.85	85.62	91.02	79.81	74.54	76.92	80.94	83.80	93.35	81.39
CDTrans	ViT-B/16	82.05	80.57	73.85	88.65	87.31	85.49	78.96	80.63	73.12	83.88	94.86	82.67
SAT-S	ViT-S/16	81.12	76.08	70.53	82.64	89.42	82.79	77.54	78.83	76.01	80.33	95.24	80.96
SAT-B	ViT-B/16	**89.77**	**85.96**	**82.64**	**90.78**	**95.50**	**89.19**	**86.74**	**84.32**	**88.28**	**87.96**	95.42	**88.78**

to assess its cross-domain performance. Based on the results in Table III, our SAT outperforms the comparative methods with significant improvement on most cross-scene tasks, reaching an average accuracy of 88.78%. Concerning task U → P, our method exhibits a slightly lower accuracy compared to the optimal methods SPDA and BA^{3}US. In this task, most methods demonstrate high performance, primarily because the source domain includes fewer outlier classes compared to other tasks. It is noteworthy that several closed-set DA methods exhibit reduced cross-domain performance in the PDA scenario. Through the comparison of the identical method in closed-set DA and PDA tasks, both CDTrans and SSRT-B exhibit a significant decrease in average accuracy. Specifically, the average accuracies drop to 82.67% and 82.74% from 91.62% and 94.40%, respectively. On the one hand, this is attributed to their ineffectiveness in handling negative transfers resulting from outlier classes, leading to misalignment. On the other hand, PDA tasks contain a considerable proportion of outlier classes within the source class space. This presents significant challenges for positive knowledge transfer across domains. The results of ViT-based methods are promising, demonstrating their robustness in handling multimodal data. Methods such as AR, BA^{3}US, and I-DANN still exhibit competitive performance, highlighting the critical significance of effectively managing outlier classes in the PDA scenario. Through our enhancements, the SAT demonstrates improved generalization ability while also effectively isolating outlier classes.

3) Analysis of Computational Complexity: We compare the parameters and computational complexity with different DA methods, and the results are presented in Table IV. It is obvious that the ViT-based methods require a significantly larger computational budget compared to those based on CNNs. The TVT exhibits the largest number of parameters and computational complexity, primarily due to its utilization of extra patch-level discriminators of multiple patches. While our method shows a slight increase in parameters and FLOPs compared to both the Baseline-B and SSRT-B, it achieves superior cross-domain performance.

TABLE IV
COMPARATIVE ANALYSIS OF COMPUTATIONAL COMPLEXITY IN DIFFERENT METHODS

Methods	Backbone	FLOPs	Params
BA^{3}US	ResNet50	16.42G	25.35M
AR	ResNet50	16.41G	24.04M
Baseline-B	ViT-B/16	33.73G	88.50M
SSRT-B	ViT-B/16	33.73G	89.48M
TVT	ViT-B/16	36.52G	98.59M
SAT-B	ViT-B/16	34.85G	90.81M

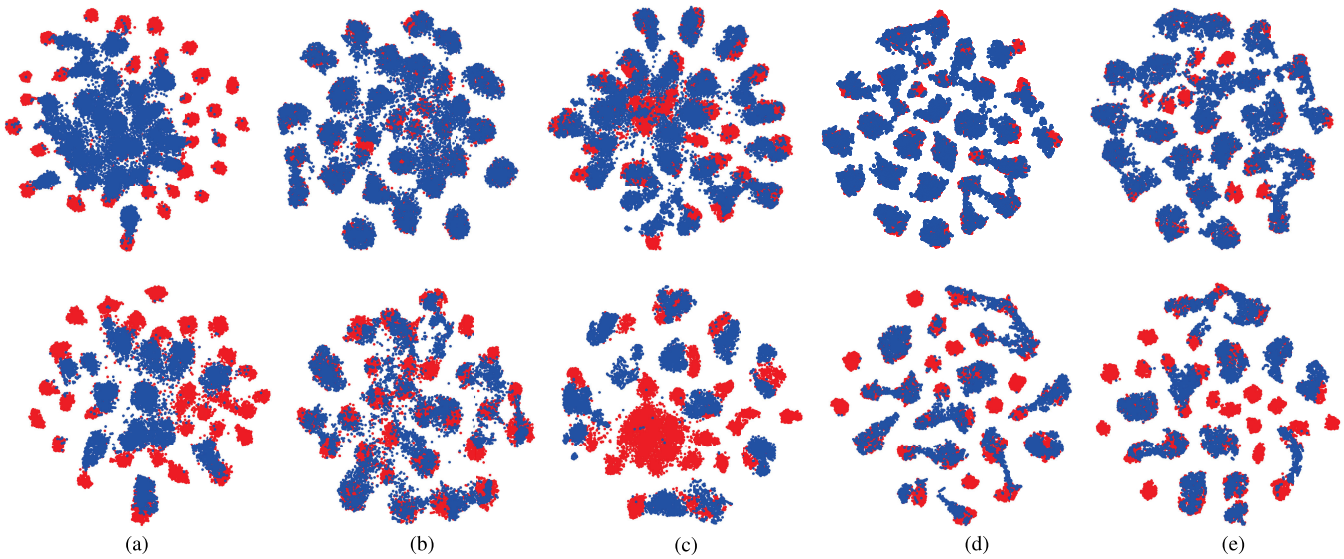
D. Ablation Studies

We present the ablation analysis to evaluate the effectiveness of the PMMD loss and reweighting strategy. The modules undergoing ablation are eliminated separately. The methods of Baseline-S/B employ the vanilla ViT-S/B with GRL for distribution alignment. This configuration also represents the result of our method without employing any reweighting strategy. Referring to the results in Table V, it is apparent that the cross-domain performance of each variant suffers different levels of degradation. The variant method by eliminating PMMD results in an 82.81% average accuracy drop, indicating that our PMMD does effectively mitigate negative transfer caused by irrelevant knowledge through the evaluation of local transferability. In the variant experiments, we noticed performance decreases of 3.86% and 5.28%, respectively, after removing class and patch weights that are denoted as CW and PW. When both of these terms are removed, the performance is inferior to that of the naive Baseline-B. Furthermore, our method can obtain the improvement of 1.88% and 2.45% by incorporating $\mathcal{L}_{\text{self}}$ and $\mathcal{L}_c$, respectively. These also validate

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TABLE V
ABLATION STUDIES FOR PDA IN CROSS-SCENE CLASSIFICATION TASKS IN RSIS

Methods	PMMD	CW	PW	Datasets											
				A → N	A → P	A → U	N → A	N → P	N → U	P → A	P → N	P → U	U → A	U → P	Average
Baseline-B				77.61	78.33	75.31	84.18	90.23	79.52	68.85	72.85	75.06	82.35	94.85	79.92
w/o PMMD		✓		83.22	85.39	79.12	87.60	93.86	83.79	76.58	72.10	79.66	75.87	93.75	82.81
w/o CW and PW	✓			75.80	76.42	74.58	86.63	91.49	78.65	70.75	70.08	73.44	82.48	94.67	79.45
w/o CW	✓		✓	85.30	80.81	80.46	88.91	93.48	85.06	79.57	79.09	84.02	82.68	94.73	84.92
w/o PW	✓	✓		79.16	79.78	77.95	88.11	92.33	83.61	76.99	78.34	83.50	82.17	94.33	83.30
w/o $\mathcal{L}_{self}$	✓	✓	✓	86.59	85.33	79.76	89.69	94.23	87.19	84.74	82.32	85.28	85.96	94.85	86.90
w/o $\mathcal{L}_c$	✓	✓	✓	87.68	**86.10**	78.27	87.86	94.13	85.99	81.32	83.45	86.04	84.26	94.54	86.33
Baselin-B+MDD		✓		79.97	76.09	76.14	86.74	91.08	79.92	74.57	77.49	82.66	82.98	94.09	81.98
SAT-B	✓	✓	✓	**89.77**	85.96	**82.64**	**90.78**	**95.50**	**89.19**	**86.74**	**84.32**	**88.28**	**87.96**	**95.42**	**88.78**



(a) (b) (c) (d) (e)

Fig. 4. t-SNE visualization results of embedded distributions on (a) source-only (ResNet50), (b) DANN, (c) BA^{3}US, (d) Baseline-B, and (e) our SAT-B, respectively. The first row and the second row display the visualization results of various methods for the tasks on A → N and A → P.

the rationality and effectiveness of the components in this real-world scenario. Our proposed method attains the highest accuracy by concurrently utilizing both mechanisms.

To further validate the superiority of our PMMD loss, we conduct an evaluation of cross-domain performance in Table V, utilizing various distribution alignment strategies on image patches. The "MDD" means that multiple domain discriminators are employed for each image patch to perform adversarial distribution alignment [29]. We observe that the performance is inferior to our method. This is because this method encounters challenges in evaluating the transferability of patches in the PDA scenario of RSIs. It equally focuses on the feature representations of all regions and fails to filter out irrelevant knowledge, ultimately causing the amplification of negative transfer and harm to distribution alignment. The results further illustrate the robustness and applicability of SAT in complex remote sensing scenarios.

E. Feature Visualization

In Fig. 4, we depict the t-SNE visualization of the feature embeddings for Source-Only, DANN, BA^{3}US, Baseline-B, and our method for PDA scenario (task: A → N and A → P). Data extracted from the source domain are visually indicated by red dots, while the blue dots illustrate data from the target domain. Our method demonstrates clear category boundaries and separability, and the source outlier classes and the shared classes are effectively isolated. In Fig. 4(b), the feature embeddings learned by DANN do not cluster as distinctly as former methods, and some outlier classes are confused with shared classes. From the results in Fig. 4(e), it is evident that the distinction between the outlier classes and shared classes is quite clear. This suggests the superiority of SAT in filtering outlier classes of source domain compared to other methods, thus effectively addressing negative transfer.

In addition, we also present the t-SNE of the embeddings at different training stages for PDA task P → N in Fig. 5. First, in the early training phase, only the source data are well-distinguished due to the label supervision. The difference in the data distribution suggests that significant domain shift still exists across domains. With the advancement of training, the embeddings of the source and target data become increasingly similar. The distributions of data form different domains

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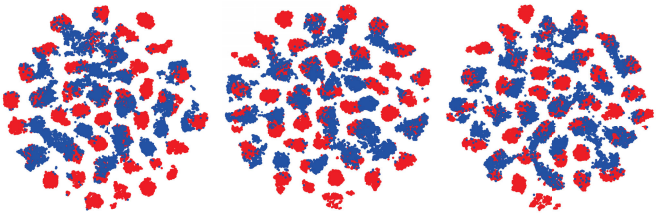


Fig. 5. Visualizations of embedding distribution during the training process by t-SNE in PDA task P → N. (a) Iteration 2000. (b) Iteration 5000. (c) Iteration 10 000.

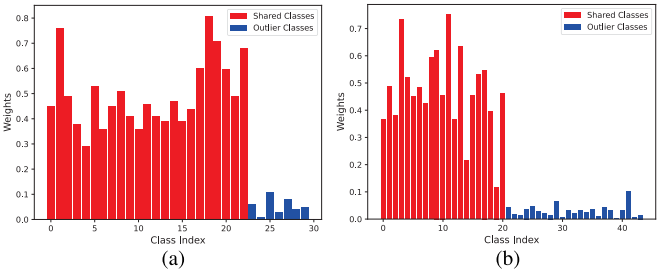


Fig. 6. Estimated class-wise weights of source domain data for the different PDA tasks. (a) A → N. (b) N → U.

are gradually grouped based on their categories, while the samples of source outlier classes are effectively distinguished from the shared classes.

These observations suggest that the SAT successfully addresses the domain shift by patch-level distribution alignment guided by the proposed bi-level reweighting strategy. The effectiveness of SAT is proven by the clustering of feature embeddings from different domains, ultimately enhancing the generalization ability of the model to tackle PDA challenges in RSIs.

F. Class Weight Visualization

As mentioned previously, the reweighting strategy is essential in PDA. To evaluate the effectiveness of the reweighting strategy, we present the class-wise weights of source examples in Fig. 6 on the PDA tasks A → N and N → U. Our method demonstrates the capability to allocate large transferable weights to classes within the shared class space $\mathcal{C}$ while simultaneously filtering out outlier classes in $\bar{\mathcal{C}}_s$ that exhibits low transferability. This facilitates the identification of source classes that closely match the target domain, thus preventing the impact of samples from outlier classes and negative transfer.

G. Proxy A-Distance for PDA

A-distance is usually employed to quantify the distribution discrepancies of the learned feature representations across domains [59]. A large A-distance implies quite significant distribution discrepancies among samples across domains, indicating that the model may not generalize well to the target domain. To verify the transferability and the generalization ability of learned features, we calculate the proxy A-distance of the source-only, DANN, SPDA, Baseline-B, and our SAT

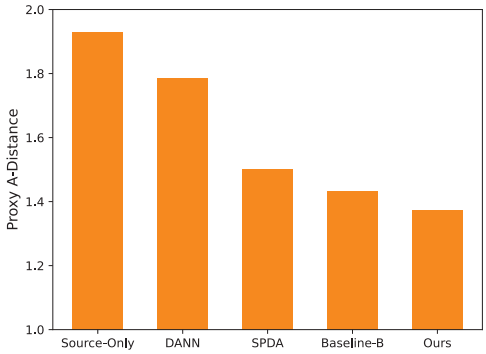


Fig. 7. Results of A-distance on PDA task A → N.

for task A → N, as represented in Fig. 7. Our method demonstrates the lowest A-distance compared to these comparative methods, indicating effective distribution alignment across datasets. Furthermore, the results also demonstrate that our method is effective in learningsrich transferable knowledge, which leads to superior cross-domain performance.

H. Analysis of Parameter Sensitivity

We discuss the parameter sensitivity on the hyperparameters λ, φ, and γ mentioned in our proposed method. In fact, it is unrealistic to achieve optimal performance across all tasks with the same configuration for parameters. Therefore, we evaluate the performance of SAT on four tasks A → N, N → P, P → U, and U → A. The results of these tasks in the PDA setting are presented in Fig. 8.

1) Parameter λ for $\mathcal{L}_{PMMD}$: The parameter λ controls the emphasis on the patch-level features' adaptation by PMMD. From the results, as shown in Fig. 8(a), the impacts of λ vary across different tasks. The performance does not show significant changes on the task N → P, suggesting low sensitivity of this task to parameters λ. The performances on other tasks show a gradual improvement as the parameter λ increases, indicating the effectiveness of PMMD in selectively aligning distributions of image patches. The overall optimal performance is obtained when λ is set to 1. Then, the same setting of λ is utilized in the following experiments.

2) Parameter φ for $\mathcal{L}_{self}$: We investigate the sensitivity by varying the parameter φ among the candidates of $\{0.01, 0.05, 0.09, 0.1, 0.5\}$. The experimental results in Fig 8(b) illustrate that optimal performance is obtained in the range of $[0.09, 0.1]$. In addition, the performances will decrease with a large value of φ, which can be attributed to reduced uncertainty resulting in overconfident mismatch. Therefore, we chose a moderately suitable value of 0.09.

3) Parameter γ for $\mathcal{L}_c$: In the PMMD, we introduce the consistency constraint between the attentive weights queried by class token and domain token. Based on the results as presented in Fig. 8(c), we set the value of parameter γ to 0.1. When the value of γ is set too high, it may result in a drop in overall performance. This could be because the flexibility in evaluating the transferability of patches is restricted, thereby preventing the model from effectively transfer knowledge.

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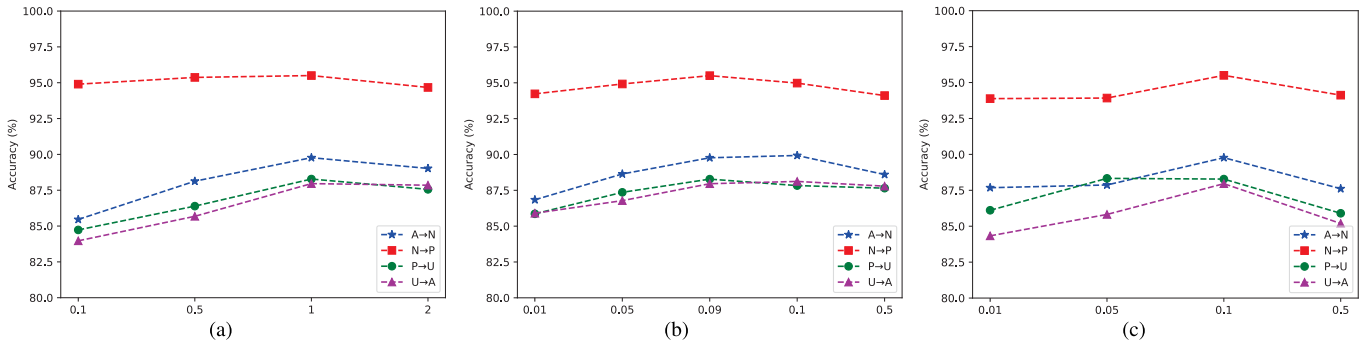


Fig. 8. Parameter sensitivity analysis of parameters (a) λ, (b) φ, and (c) γ on PDA tasks A $\rightarrow$ N, N $\rightarrow$ P, P $\rightarrow$ U, and U $\rightarrow$ A.

V. Conclusion

In this article, we propose a new SAT for RSIs cross-scene classification tasks under the PDA scenario. Different from previous PDA methods that allocate an equal weight to entire samples without filtering irrelevant knowledge, the proposed SAT selectively aligns the distributions from both class- and patch-level perspectives. Specifically, we introduce a bi-level reweighting strategy that can further evaluate the transferability of image patches. Furthermore, we propose a customized PMMD loss to selectively align transferable image patches that are more likely to benefit the distribution adaptation while preventing the untransferable parts. By doing this, we enhance the positive knowledge transfer while mitigating the negative transfer from source outlier classes. Experimental results demonstrate that SAT successfully tackles the PDA problem in RSIs and outperforms other comparative methods. Moreover, our proposed SAT is effective, which addresses the unique challenges for the cross-scene classification task in RSIs under the PDA setting. This also underscores the significance of considering the characteristics of RSIs in real-world applications. Our future research involves addressing challenging tasks, such as domain generalization and its applications in RSIs.

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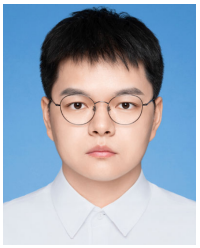
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